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Carbodicarbene Monolayers on Au(111) Enable Strong Carbon(0)-Metal Binding and Work-Function Modulation

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ABSTRACT

Carbodicarbenes (CDCs), featuring a formally zero-valent carbon atom coordinated by two N-heterocyclic carbene (NHC) units, are exceptionally strong σ-donor ligands in molecular chemistry, yet their ability to functionalize and electronically modulate solid surfaces has remained unexplored. Here, we introduce CDCs as a new class of molecular surface ligands for nanoscale interface engineering and demonstrate their assembly into robust monolayers on Au(111). Scanning tunneling microscopy (STM) imaging, supported by simulations, resolve the nanoscale organization and adsorption geometry of the CDC monolayers, showing that CDCs adsorb predominantly in a tilted configuration and transform upon annealing into a flat-lying geometry without long-range surface diffusion. This behavior is consistent with strong ligand-surface coupling and a high degree of interfacial stability. X-ray photoelectron spectroscopy (XPS) and polarized near-edge X-ray absorption fine structure (NEXAFS) measurements further confirm the formation of chemically anchored, thermally stable CDC monolayers and the thermally-induced shift in their adsorption geometry. Ultraviolet photoelectron spectroscopy (UPS) reveals that CDC adsorption decreases the Au work function by up to 1 eV, demonstrating substantial electronic modulation of the metal surface. These experimental findings, supported by Density functional theory (DFT) calculations, establish CDC as a surface- ligand for molecularly controlled work-function modulation and interface engineering.

1 | Introduction

CDCs represent a distinctive class of carbones in which the central carbon atom, coordinated to two NHC units, adopts a formally zero-valent state (Figure 1a) [1–4]. It was identified that in the resonance structure of CDC, the central carbon possesses two orthogonal lone pairs of electrons, enabling it to function as both a strong σ-donor and π-donor [1, 5]. This characteristic distinguishes CDC from traditional NHC-type singlet carbenes, which are strong σ-donors and weak π-acceptors, and positions CDC as a unique ligand with exceptional affinity to complexation with metals [3, 6]. The strong donor character of CDCs originates from the filled p-type orbitals on the central carbon atom, in contrast to NHC which have an empty p-orbital [7–9]. CDCs function as strong σ-donors that outperform NHCs and phosphines, as

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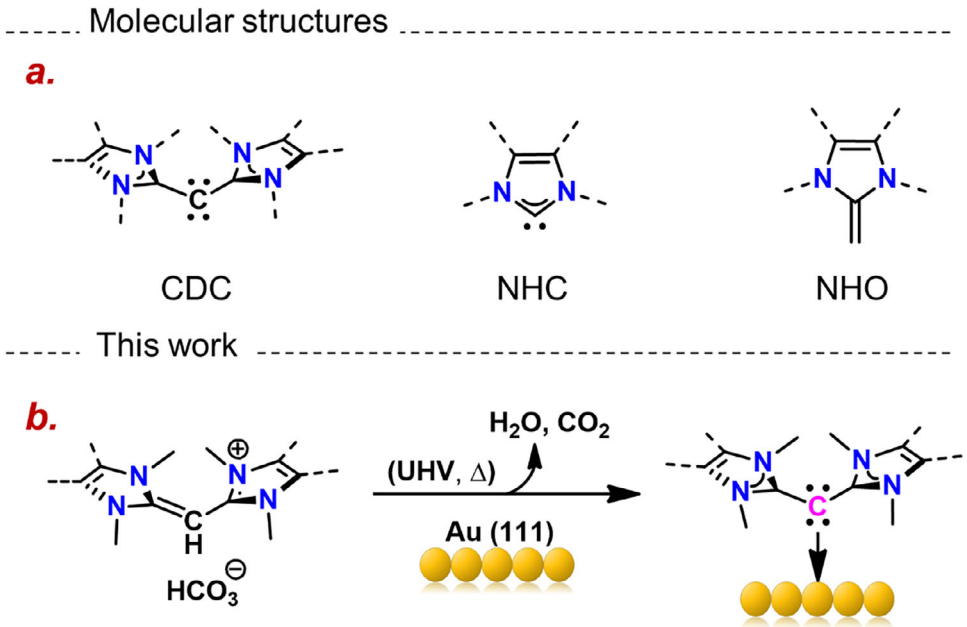


FIGURE 1 | (a) Molecular structures of carbodicarbene (CDC), N-heterocyclic carbene (NHC) and N-heterocyclic olefin (NHO). (b) CDC deposition procedure under UHV conditions.

validated by both DFT calculations [10] and experimental results [2, 11].

As a result of their unique electronic properties, carbones have emerged as a distinct class of ligands with considerable utility in alkali earth metals [12, 13], transition metals [14–17], main group chemistry [18–20] and nanoclusters [21]. CDCs have been shown to enable diverse reactivity, including Frustrated Lewis Pair (FLP) functionality [22–24], ligand exchange [25–27], and catalytic transformations [28]. CDCs exhibit enhanced ligand efficiency compared to N-heterocyclic carbenes (NHCs), effectively stabilizing main-group elements like boron [20, 29] and bismuth [30].

It was previously demonstrated that NHCs [31–37] and N-heterocyclic olefins (NHOs) [38–41], which construct the subunits of CDC, can function as surface ligands. The strong σ-donation affinity of NHCs and NHOs makes them highly robust surface ligands on various metal and metal-oxide substrates [42–44]. Their strong interaction with surface atoms facilitated their self-assembly into monolayers with enhanced thermal and chemical stability, compared to thiols [32, 45]. The synthetic versatility and strong surface affinity of NHCs and NHOs led to their integration as surface ligands in various applications [46, 47], for example, as (bio)sensors [48–50], ligands on nanoparticles and clusters [51–59], molecular probes for tuning and analysis of surface reactivity [60–62], electronic device modifiers [63–68], co-catalysts [69–72], and corrosion inhibitors [73–76].

The strong affinity of CDCs to metal ions highlights their potential as unique surface ligands on metal films for modulating the electronic properties in ways that may surpass those of NHC and NHO ligands. However, the functionality of CDCs as surface ligands has not yet been explored. This presents a compelling opportunity to expand the application space of CDCs beyond coordination chemistry into surface science and materials design. Herein, we investigate the self-assembly of CDCs on Au(111) under ultra-high vacuum (UHV) conditions (Figure 1b) and analyze the CDC monolayer impact on the electronic properties of the Au surface. X-ray spectroscopy measurements, coupled with DFT calculations, show that the electron-rich C(0) center of CDC establishes a strong σ-bond with the surface. The high thermal stability of CDC ligands is identified by X-ray absorption spectroscopy and scanning tunneling microscopy (STM) measurements, which also detect a transition in the adsorption geometry of CDC from tilted to flat-lying geometry upon annealing. In addition to its enhanced thermal stability, the CDC monolayer significantly lower the work function of the Au surface by up to 1 eV, attributed to ligand-surface charge transfer, which is also identified in DFT calculations. These results demonstrate the dominant impact of strong CDC-Au interactions on the electronic properties of the Au surface.

2 | Results and Discussion

In this work, we explore the anchoring of CDC-1 and CDC-2 (Figure 2) on Au(111) and the impact of CDC anchoring on the electronic properties of the Au surface. CDC-1 and CDC-2 differ in their backbone functionalization and were selected for this study to examine the effect of CDC backbone functionalization on the self-assembly. In order to generate the desired CDC we used bench-stable hydrogen carbonate salts $1{\cdot}HCO_3$ and $2{\cdot}HCO_3$ as precursors (Figure 2), a procedure established for the generation of carbenes via sublimiation [54, 48, 77–81]. It should be noted that CDC formation and surface-binding do not necessarily require an inert atmosphere and can also be prepared under atmospheric conditions, as will be described further below.

$1{\cdot}HCO_3$ and $2{\cdot}HCO_3$ were synthesized and characterized (detailed synthetic procedures and characterization data are provided in Table S1, and Figures S1–S15). Surface ligands were

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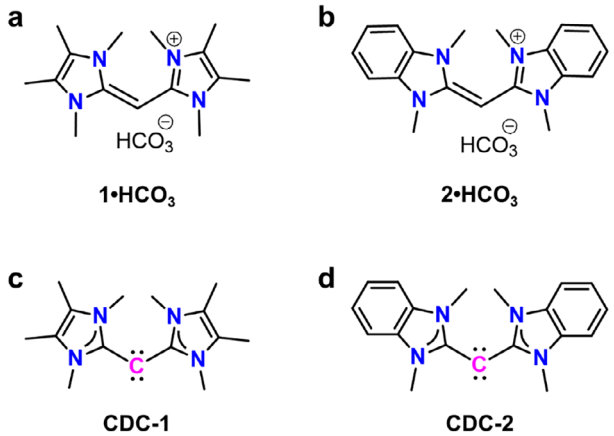


FIGURE 2 | Molecular structures of the bicarbonate salts (a) $1{\cdot}HCO_3$ and (b) $2{\cdot}HCO_3$ used as precursors for the formation of surface ligands, (c) CDC-1 and (d) CDC-2, respectively, via annealing and sublimation under UHV conditions.

prepared by sublimation of the corresponding bicarbonate salts toward the Au(111) substrate under UHV conditions, which resulted in CDC formation and surface anchoring. The deposition was carried out via in-vacuo sublimation of the solid precursor at 364 K for CDC-1 and 374 K for CDC-2 using a boron nitride crucible, resulting a typical base pressure of $3{\cdot}10^{-8}$ mbar [48, 80]. The CDC surface ligands were characterized using X-ray photoelectron spectroscopy (XPS) and polarized near-edge X-ray absorption fine structure (NEXAFS) spectroscopy, carried out at the ALOISA beamline of the ELETTRA synchrotron (Trieste, Italy) [82, 83].

N1s XPS signals of CDC-1 and CDC-2 were acquired at room temperature following their adsorption on Au(111) and are shown in Figure 3a(i),b(i). The N1s spectra displayed a broad and asymmetric structure peaked at 400.8 and 401.0 eV for CDC-1 and CDC-2, respectively. A shoulder at lower binding energy (~399 eV) was identified for both CDCs and assigned to physisorbed residues, as identified by the decrease of this signal upon mild annealing to 150°C (Figure 3a(ii),b(ii)).

For comparison, the photoemission spectra of the corresponding carbonate species ($1{\cdot}HCO_3$ and $2{\cdot}HCO_3$), as measured in the powder state, displays the main peak (largely dominant) at binding energy of 400.0 and 399.8 eV, respectively (Figure S16). The deprotonation following surface anchoring thus determines a core level shift of ~1 eV for both CDC-1 and CDC-2. Similar shifts in the N1s peak position due to deprotonation and self-assembly were reported for NHC and NHO that were prepared by a similar approach [38].

The asymmetric and relatively broad (~2 eV) N 1s spectra observed at room temperature indicate a distribution of local electronic environments for the nitrogen atoms, which may originate from variations in adsorption geometry and/or adsorption-induced electronic asymmetry within the CDC molecules. Upon annealing, a distinct high binding-energy component at 401.5–401.6 eV develops and the two components become clearly resolved. This splitting may reflect adsorption-induced symmetry breaking of the CDC framework, resulting in electronically nonequivalent carbene-derived nitrogen environments, rather than necessarily representing two separate molecular adsorption configurations. Further annealing to 300°C (Figure 3a(iii),b(iii)) progressively increases the relative contribution of the high-binding-energy component, with the lower-binding-energy component becoming nearly absent for CDC-1.

The binding energy of the two components remains well within the upper and lower limits set by the olefinic (402.2 eV) and carbene (401.2 eV) values, which can rule out the hypothesis of CDC dissociation into its NHO and NHC subunits. According to NEXAFS measurements, along with STM measurements and DFT calculations, discussed in the following, the spectral changes may be attributed to changes in relative population of different adsorption configurations and asymmetric electronic polarization of the CDC upon adsorption on Au.

A better insight into the thermal robustness of CDC films is provided by analyzing the temperature evolution of the integral intensity of the relevant photoemission spectra. The changes in the N1s XPS peak area (Figure 4a) and the N1/C1s peak area ratio (Figure 4b) as function of the annealing temperature were calculated for CDC-1 and CDC-2 monolayers, and are marked in black and red colored circles, respectively, in Figure 4 (XPS spectra are shown in Figure S17).

Upon stepwise annealing to 300°C, the N1s XPS signal intensity of CDC-1 and CDC-2 gradually and steadily decreased by approximately 35%–40% (Figure 4a). Annealing to 150°C leads to desorption of weakly bound molecules and partial conversion into a single chemical species associated with the symmetrically splitted N peak (Figure 3a(ii),b(ii)). Annealing to 300°C resulted in a decrease in the intensity of the main feature, which is attributed to preferential desorption of CDC molecules from weaker-binding configurations, likely associated with terrace sites lacking optimal Au ad-atom coordination or metastable adsorption geometries.

The N1s/C1s peak area ratio for CDC-1 (Figure 4b) showed a decrease of ~ 20% upon annealing to 300°C. The stronger decrease in the N1s intensity relative to the C1s signal (Figure S17) suggests that partial decomposition of CDC-1 occurs in addition to molecular desorption, leading to preferential loss of N-containing fragments and retention of carbon-rich surface species. Upon annealing to 350°C the peak area ratio showed additional decrease and changes, which were detected in both the N1s and C1s XPS pattern (Figure S17), indicative of further thermal-induced decomposition.

The N1s/C1s peak-area ratio for CDC-2 increased after annealing to 200°C (Figure 4b). This primarily reflects a decrease in the intensity of the low energy component in the C1s signal (Figure S17) and may result from the preferential removal of weakly bound carbon-containing species or from thermally induced reorganization of the adsorbed layer. N1s/C1s peak area ratio of CDC-2 showed a moderate decrease of ~8% upon further annealing to 300°C. Annealing to 350°C led to a pronounced decrease in the N1s/C1s peak-area ratio, which is indicative of molecular fragmentation. Its N1s and C1s XPS binding energies shifted toward lower values (Figure S17), corresponding to decomposition and formation of a pyridine-like nitrogen species on the

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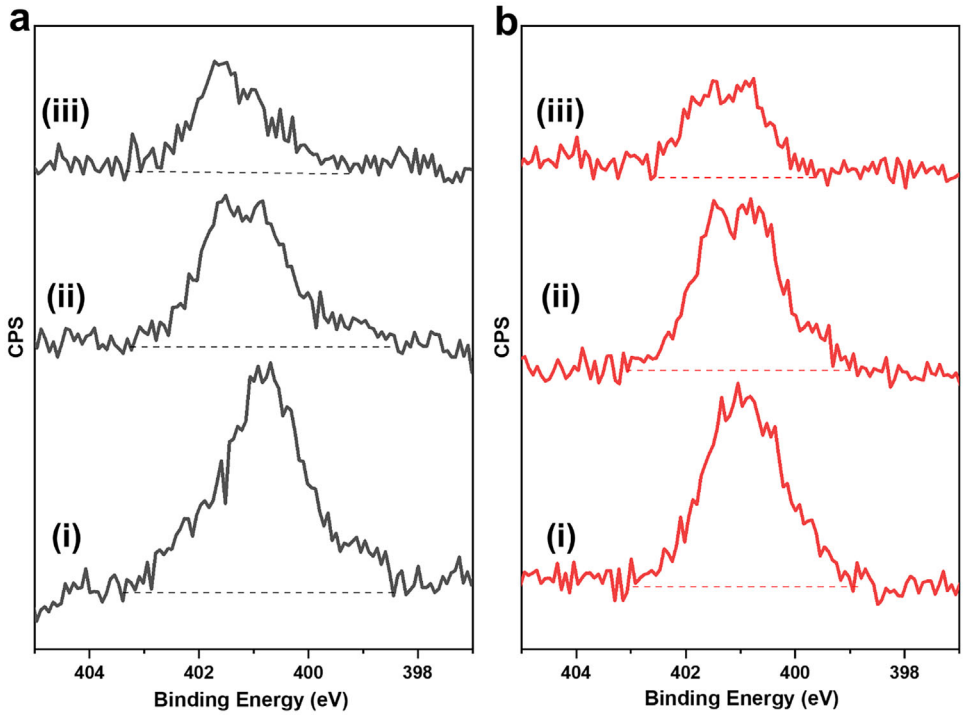


FIGURE 3 | N1s XPS signals of CDC-1 (a) and CDC-2 (b) on Au(111) at room temperature (i) and after annealing to 150°C (ii) and 300°C (iii).

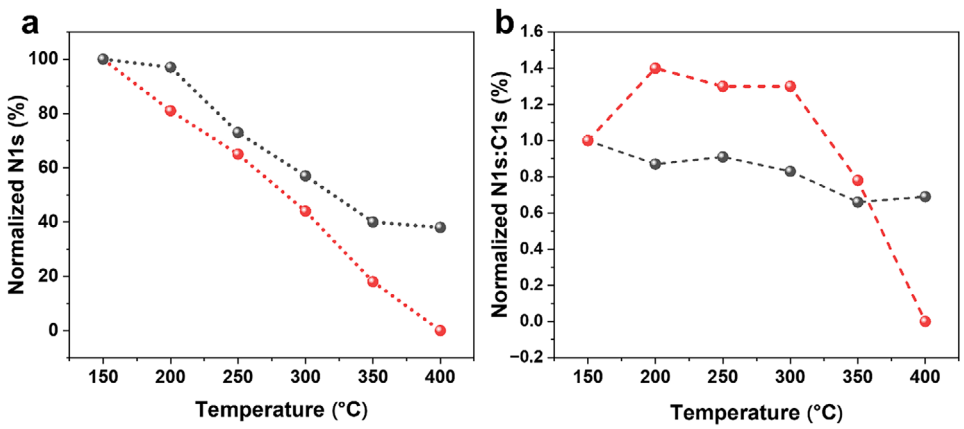


FIGURE 4 | (a) Normalized N1s XPS peak area of CDC-1 (black) and CDC-2 (red), and (b) the corresponding N1s/C1s XPS peak-area ratios for CDC-1 (black) and CDC-2 (red).

surface [84, 85]. It should be noted, however, that most CDC-derived adsorbed species remained detectable on the Au surface even after annealing to 300°C. This high thermal stability is consistent with the strong σ-donor character of CDC ligands, which promotes robust CDC-Au bonding and stabilizes the adsorbed monolayer, thereby favoring partial thermal decomposition over desorption at elevated temperatures.

Nitrogen K-edge NEXAFS measurements were conducted to probe the electronic structure and average orientation of the CDCs (Figure 5). For CDC-2, the spectrum measured at room temperature in p-polarization (dotted line in Figure 5a) displays a prominent π^* transition at 401.7 eV and a sharp satellite resonance at 403.3 eV. The first resonances matches the value reported for NHOs (401.8 eV) [38], while the second one corresponds to a higher energy molecular orbital associated with the conjugated ring, as observed for benzene-NHC on Au [86]. The absorption spectra of CDC-1 (Figure S18) displays the first sharp resonance at the same energy of CDC-2, whereas the second feature is replaced by a shoulder on the high energy side of the main one, equivalent to the absorption spectra of NHOs [38]. Overall, the energy of the first resonance of both CDCs is the same as that of NHOs and NHCs, and mostly differ by their satellite fine structure [38]. At energies above the ionization threshold, the broader spectral intensity is predominantly associated with σ^* excitations of the C—N framework.

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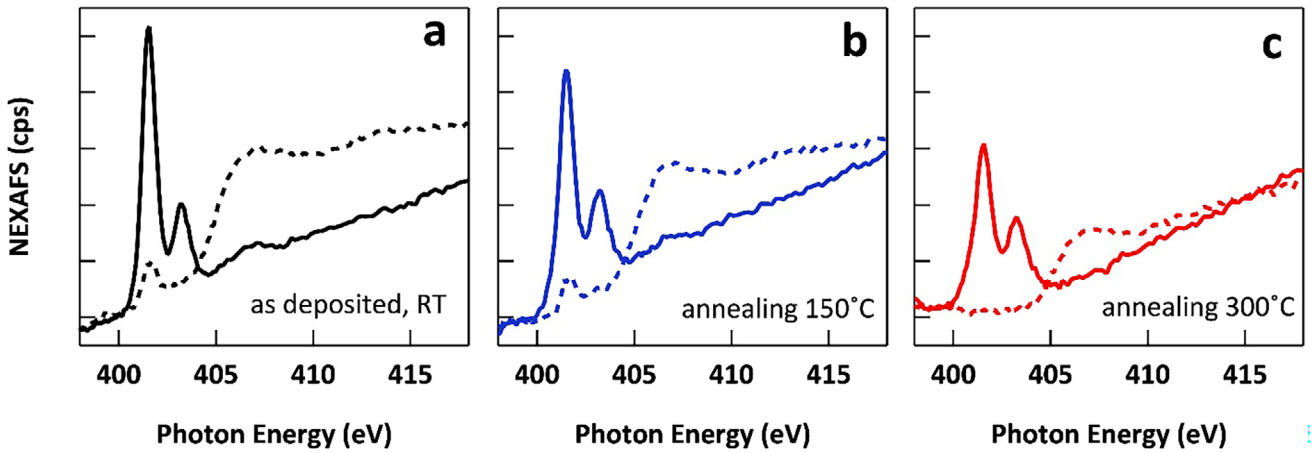


FIGURE 5 | Nitrogen K-edge NEXAFS spectra of CDC-2 measured at p- and s-polarizations (dotted and dashed lines, respectively) at room temperature (a) and after annealing to 150°C (b) and 300°C (c).

The X-ray absorption spectra measured in s-polarization display a strong enhancement of the σ^* symmetry resonances, which is above the ionization threshold, and, consistently, a decrease of the intensity of π^*-symmetry ones. The intensity ratio of the resonances in s- and p-polarization provides indications about the average orientation of the corresponding molecular orbitals with respect to the surface. The NEXAFS linear dichroism of both CDCs indicates a preferential orientation of the π^* molecular orbitals associated with the aromatic rings toward the surface. However, an unambiguous determination of the overall CDC molecular orientation is not straightforward because multiple adsorption configurations may coexist, and the molecules can adopt different equilibrium conformations, as further indicated by DFT calculations.

As the temperature increases (Figure 5b,c), the linear dichroism increases as well, pointing to a gradual flattening of the molecules on the surface. The starting configuration of CDC-1 (Figure S18) is flatter than CDC-2 (Figure 5a), but even the latter one evolves into an orientation parallel to the surface at 300°C, before CDC-2 desorption takes place. The fine structure absorption line shape of CDC-2 remains practically unaltered up to the desorption temperature, displaying only a minor broadening of the main resonances due to the closer proximity to the surface of the flat aromatic rings. This experimental evidence confirms the indications from XPS, clearly showing that the electronic structure of the molecular orbitals is thermally robust up to 300°C.

The spectra of Figure 5 are shown on an absolute intensity scale, where one may appreciate the overall intensity decrease with temperature, in good agreement with that observed in XPS. Most importantly, the energy and shape of NEXAFS resonances do not change from RT to 300°C, in addition they always show a large dichroism. These two observations indicate that i) the signal stems from N atoms embedded in the pristine molecular structure, and ii) the corresponding molecular orbitals remain practically unchanged through the different annealing stages. We observed a decrease of the intensity ratio between the leading π^* resonance at 401.7 and its main satellite at 403.3 eV, which we attribute to the conformational differences of molecules in the film as deposited at RT (multiple species), and next conformational changes at increasing temperature, as discussed in the following.

The polarization-dependent NEXAFS measurements therefore provide an ensemble-averaged picture of the CDC adsorption geometry, showing a preferentially tilted orientation at room temperature and progressive flattening of the molecular ring system upon annealing, while the persistence of the principal N K-edge resonances indicates that the nitrogen-containing molecular framework remains largely intact up to 300°C. A more detailed assignment of the specific adsorption configurations and their thermally induced interconversion requires integration of the NEXAFS data with complementary STM measurements and DFT calculations, as discussed below.

DFT calculations were conducted to determine the optimal adsorption geometries of CDCs on Au(111) surface. Since Au ad-atoms induce stronger adsorption of NHCs and NHOs on Au(111) [38, 87] and increase the ligand's mobility for the formation of an ordered monolayer [33], CDC configurations with and without an ad-atom were calculated (see experimental section for details about the calculations of ad-atom formation).

To identify the preferred adsorption geometries, a systematic exploration of plausible initial positions and molecular orientations was performed for CDC-1 and CDC-2 on Au(111) and Au ad-atom/Au(111). During geometry optimization, the molecules were allowed to relax without constraints on their tilt or rotational orientation, and multiple distinct starting configurations were tested. Several initial structures converged to the same local minima, leading to the identification of three principal adsorption motifs: Vert-Vert, Vert-Flat, and Flat-Flat. Figure 6 shows the lowest-energy configuration obtained for each adsorption motif, while additional optimized configurations and their corresponding formation energies are reported in Table S2.

Three conformers were obtained for CDC-1 (Figure 6a–c) and CDC-2 (Figure 6d–f). In the Vert-Flat configuration, the CDC-1 and CDC-2 (Figure 6a,d) are positioned with one carbene ring parallel and one perpendicular to the surface and an ad-atom is extracted from the bulk. This ad-atom is strongly interacting through the central carbene atom with a C-Au distance of ~2.05

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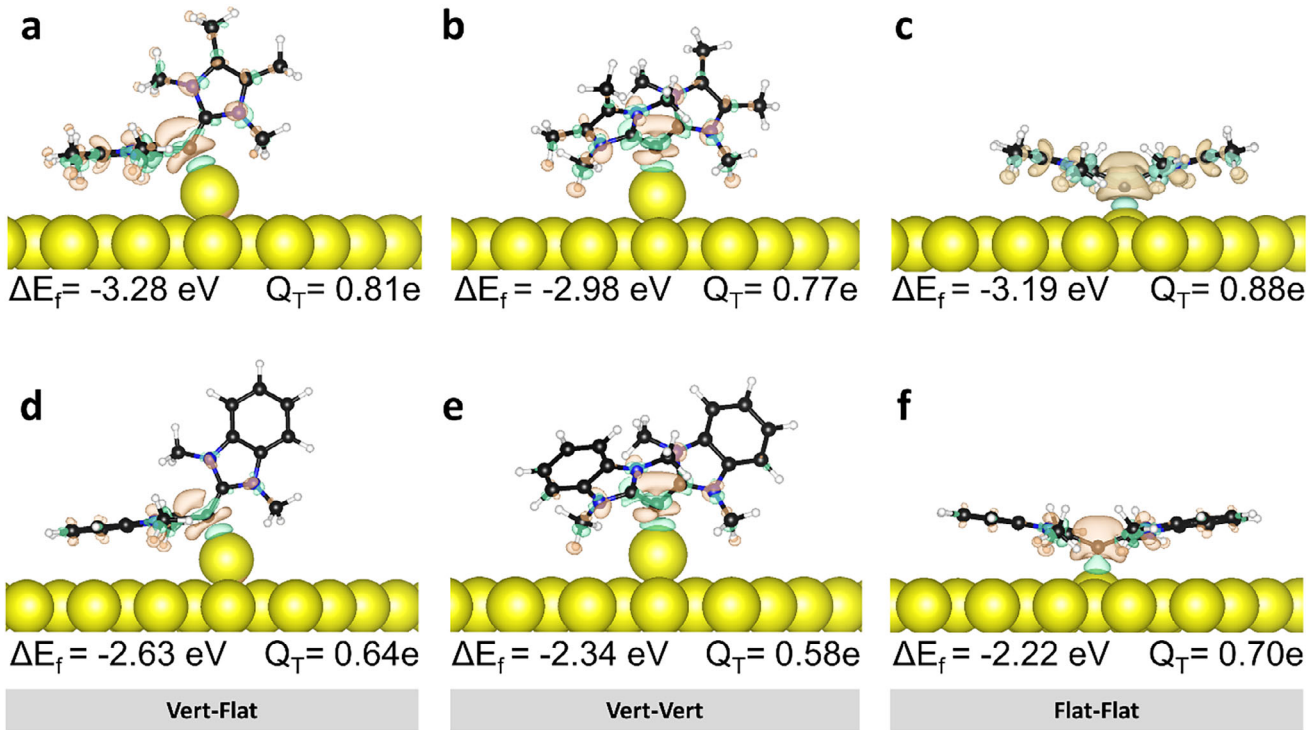


FIGURE 6 | Optimized adsorption geometries for CDC-1 (a–c) and CDC-2 (d–f) on Au surfaces. Three structures were optimized on Au ad-atom on Au(111) (a,b,d,e) and Au(111) (c,f): Vert-Flat (a and d), Vert-Vert (b and e), and Flat-Flat (c and f). The formation energy (ΔE_f) is provided in eV. The electron transfer from the substrate to the Au surface is illustrated by an iso-surface at 0.003e corresponding to the charge difference between the isolated fragments and the chemisorbed structure. The overall electronic transfer from CDC to the Au atoms is given below as Q_T and was quantified using Bader charges. Au in yellow van der walls spheres, C in black, N in blue, H in white.

Å. This is the most stable configuration for both CDC-1 and CDC-2 (−3.28 and −2.63 eV, respectively). In absence of any ad-atom, the formation energy of Vert-Flat is noticeably lower by more than 0.5 eV (−2.88 and −2.19 eV for CDC-1 and CDC-2, respectively, see Table S2). Calculations of the formation energies of the corresponding NHCs on ad-atom without and with a benzyl ring yielded formation energy of −2.32 and −2.14 eV, respectively. These results indicate that, relative to NHCs, CDCs exhibit stronger adsorption affinity to the Au surface by ~0.4-0.9 eV.

The second structure of chemisorbed CDCs, termed Vert-Vert, shows two carbene rings at a tilted angle to the surface (Figure 6b,e for CDC-1 and CDC-2, respectively), one with an angle to the surface normal around 20° and the other one around 47°. In this adsorption geometry the central carbene atom is strongly interacting with an ad-atom. Last, a flat-lying chemisorption mode in which both carbene rings are parallel to the surface, termed Flat-Flat, was identified for the two CDCs (Figure 6c,f for CDC-1 and CDC-2, respectively). In this adsorption geometry the central carbene atom in the CDCs strongly binds to one of the Au surface atoms, which is positioned slightly above the surface. For CDC-1, a formation energy of −3.19 eV and a C-Au distance of 2.12 Å were identified. In the case of CDC-2, the formation energy was −2.22 eV and the C-Au distance was slightly longer (2.16 Å). The variance in the formation energy between CDC-1 and CDC-2 is correlated to a larger distortion cost of CDC-2, compared to CDC-1, that can be quantified by computing the deformation energy (energy required to distort the surface and the molecule into the adsorption geometry), as reported in Table S3. For both CDCs the adsorption on an ad-atom was found less favorable for the Flat-Flat structure (Table S2). The Flat-Flat adsorption mode is higher in energy than the Vert-Flat adsorption mode on Au ad-atom, but for both CDCs becomes a favorable geometry on Au(111), as shown in Table S2.

STM imaging at low surface coverage was performed for CDC-2 to provide a detailed analysis of the adsorption geometry (Figure 7). CDC-2 was chosen for this analysis due to its higher on-surface stability, as will be discussed below. At low surface density of CDC, nonsymmetric structures were detected at the elbow sites of the herringbone reconstructed surface (Figure 7a,b), along with a lower density of structures exhibiting a dimmer signature. Higher surface coverage of CDC-2 did not yield an ordered molecular layer (Figure S19) and no formation of dimers or trimers, which were previously probed for NHCs [42, 88] and NHOs [38], were detected for CDCs.

The brighter structures did not change their pattern upon annealing to 150°C (Figure 7c,d), but the dimmer structures were not probed after annealing, which is indicative that these are physisorbed species. The desorption of physisorbed molecules following annealing was also identified by XPS measurements (Figure 3). Annealing to 200°C led to small changes in the structure of the molecules but did not change their distribution,

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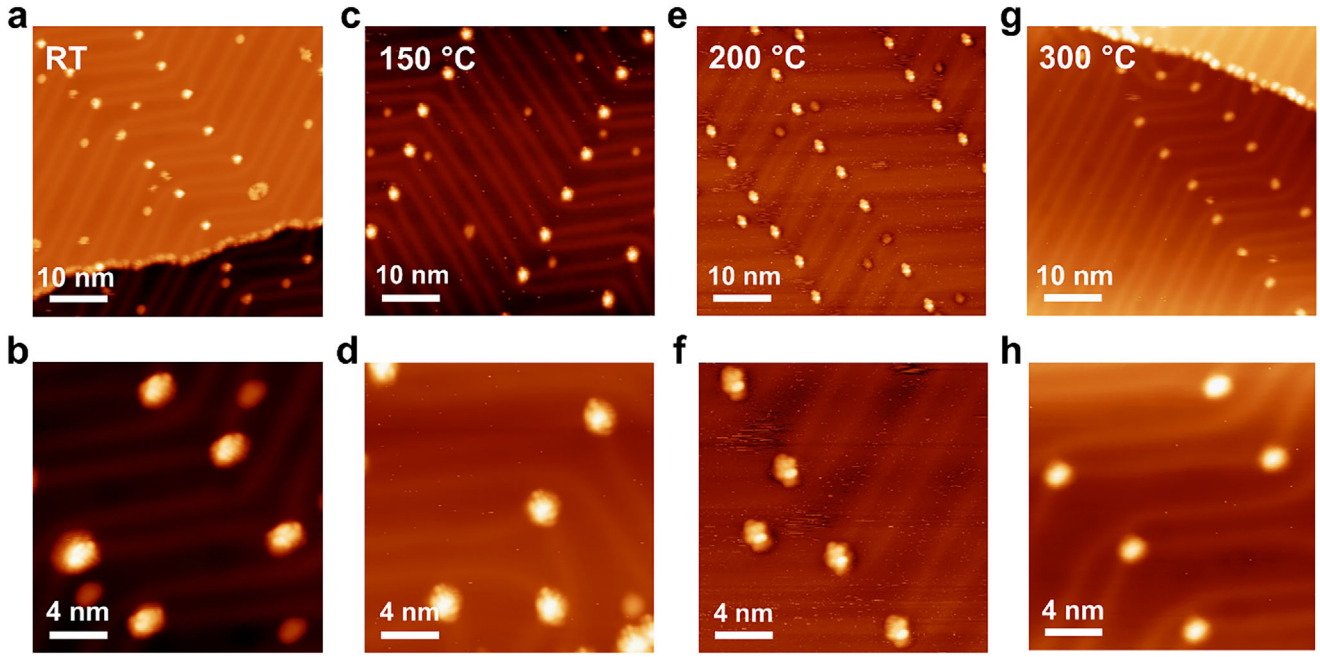


FIGURE 7 | (a) STM image of CDC-2 deposition on Au(111) at room-temperature (acquisition parameters: −1.3 V, 80 pA). (b) High-resolution image of (a) (acquisition parameters: −1.3 V, 100 pA); (c) After annealing to 150°C (acquisition parameters: −1.3 V, 70 pA). (d) High-resolution image of (c) (acquisition parameters −1.3 V, 70 pA). (e) After annealing to 200°C (acquisition parameters: −1.5 V, 60 pA). (f) High-resolution image of (e) (acquisition parameters −1.5 V, 60 pA). (g) After annealing to 300°C (acquisition parameters: −1 V, 50 pA and (h) High-resolution image of (g) (acquisition parameters: −1 V, 50 pA). STM imaging was conducted at a liquid nitrogen temperature (78 K).

indicating that these molecules are chemisorbed on the Au surface (Figure 7e,f).

STM images of the stable structures along with their relative height profiles are shown in Figure 8. The asymmetric structure that was identified as the dominant species at 150°C was characterized with a relative height of 0.45 nm (Figure 8a,b). Annealing to 200°C led to sharpening of the asymmetric structure (Figure 8c,d). The probed structure was correlated to the highly stable Vert-Flat geometry (Figure 6d). STM Simulations of CDC-2 at a Vert-Flat geometry were performed and resulted an asymmetric pattern (Figure S20a), which is very similar to the one experimentally probed by STM.

STM images of CDC after annealing to 300°C show no change in the surface density and distribution of CDCs (Figure 7g,h), but their structure was modified into a symmetric pattern. Height profile analysis showed that the symmetric structure is flatter than the asymmetric one (Figure 8e,f). The simulated STM signatures further distinguish between the two Flat-Flat adsorption configurations. The Flat-Flat structure bound to Au ad-atom produces a relatively elongated and asymmetric feature (Figure S20b), whereas adsorption directly on Au(111) results in a more compact, approximately oval feature with a more pronounced apparent twofold symmetry (Figure S20c). The latter is more consistent with the increased symmetry of the experimentally observed CDC-2 structures following annealing to 300°C.

It is hypothesized that exposure to elevated temperature led to mobility of surface atoms and the structural changes in the Au surface favored the transition to the Flat-Flat geometry on Au(111). An energetically accessible pathway from the Vert-Flat to the Flat-Flat configuration involves initial reincorporation of the Au ad-atom into the Au surface while retaining the Vert-Flat molecular geometry (+0.11 eV for CDC1 and +0.44 eV for CDC2) and then reorientation from Vert-Flat to Flat-Flat on pristine Au(111) (−0.32 eV for CDC1 and −0.03 eV for CDC2). The reincorporation of Au ad-atom on defect sites will be even lower in energy and thus more feasible. In addition, the decrease in the molecular coverage upon annealing may alter lateral CDC-CDC interactions and surface packing, thereby favoring the Flat-Flat geometry at elevated temperature. CDC-2 adsorption on Au dimer is less favorable than adsorption on a single Au ad-atom by 0.35 eV, which is correlated with a less optimal electronic interaction (Figure S20).

STM imaging of CDC-1 on Au(111) revealed pronounced tip-induced molecular mobility at sub-monolayer coverage, which prevented stable intramolecular resolution and therefore limited a direct structural analysis of its adsorption geometry (Figure S21). It can be hypothesized that the high on-surface mobility is correlated to the stronger CDC-1 interaction with the Au ad-atom, which facilitates the on-surface diffusion [33].

Integration of the XPS, NEXAFS, STM and DFT data leads to the conclusion that CDC can be positioned either at Vert-Flat and Flat-Flat positions in which the nitrogen atoms differ in their electronic environments. Annealing changed the distribution between these geometries. Annealing to 150°C led to desorption of the physisorbed species and the CDC favored the Vert-Flat position. Annealing to 300°C led to additional changes in the adsorption geometry, as obtained by XPS, NEXAFS and STM measurements, with preference to a Flat-Flat adsorption geometry. The Flat-Flat adsorption mode is higher in energy (+0.41 eV) than

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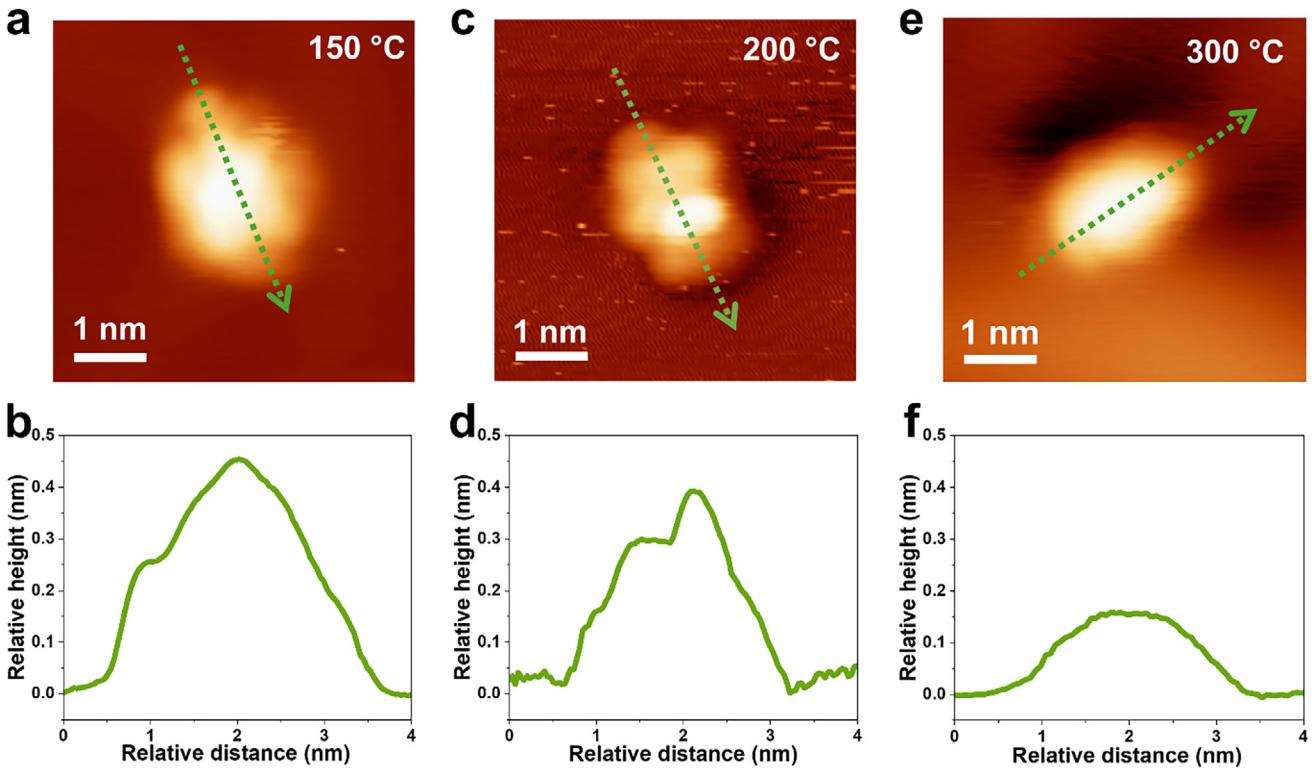


FIGURE 8 | STM imaging of CDC-2 along with their corresponding relative height profiles after annealing to 150°C (a and b, respectively), 200°C (c and d, respectively) and 300°C (e and f, respectively).

the Vert-Flat adsorption mode on Au ad-atom, but for both CDCs becomes a favorable geometry on Au(111), as shown in Table S2.

NEXAFS dichroism of the as-deposited monolayer (Figure 5) yields an intensity ratio in the s- and p-polarization of ~20%. By bare application of the symmetry rules [89] to the ALOISA scattering geometry, such a value would correspond to an average tilt angle of ~30° from the surface for a simple coplanar CDC conformation (Figure S22). If we rather assume a Vert-Flat structural model, one aromatic subunit (down) is always parallel to the surface, thus the measured intensity in s-polarization only stems from the single aromatic subunit (up) tilted off the surface. Assuming that both up and down subunits yield the same x-ray absorption cross-section, then a linear dichroism of 20% yields an average tilt of 50° for the up subunit in the ALOISA scattering geometry. This value largely reduces the discrepancy between the measured NEXAFS dichroism and the expected average tilt of the DFT model for the up subunit (~70°). Keeping into account the XPS indication of multiple molecular configurations, we may conclude that the population of Vert-Flat molecules is the majority one in the as-deposited film. The combined effect of partial thermal desorption and conformational change to Flat-Flat geometry leads to a CDC film made of a population dominated by molecules with their conjugated rings oriented approximately parallel to the surface upon annealing to 300°C.

DFT analysis of the electronic charge of CDCs was conducted to quantify the electronic properties of CDCs following their chemisorption. This analysis was performed by combining the total Bader charge transfer (Figure 6), atomic Bader charges (Figures S23 and S24) and the electronic localization function (ELF) analysis computed on the isolated and chemisorbed CDCs (Figure S25). ELF analysis of the gas phase CDC clearly indicates the presence of a lone pair of electrons on the central carbon. Upon chemisorption, this lone pair interacts with the Au surface. Charge analysis shows that changes in the adsorption geometry modify the charge transfer to the Au surface (Q_T values in Figure 6). Equivalent NHCs (Figures S26–S28) showed lower charge transfer values, which is correlated to their weaker adsorption energy, as was analyzed for dimethyl-NHC (ΔE_f = −2.32 eV, Q_T = 0.41e) and dimethyl-benzNHC (ΔE_f = −2.14 eV, Q_T = 0.37e). The isosurface of the charge difference (Figure 6) shows that the main atoms involved in the charge flow are the central carbene atom and the surrounding NCN motifs.

Ultraviolet photoelectron spectroscopy (UPS) measurements were conducted to probe the influence of CDC monolayer on the electronic properties of Au(111) (Figure 9). For these measurements the deprotonation of 1·HCO_3 and 2·HCO_3 precursors were conducted under liquid-phase conditions (see supporting information for details). The surface deposition of CDCs by this approach was validated by N1s XPS analysis and laser desorption ionization mass-spectrometry (LDI-MS) measurements (Figures S29 and S30). Polarization modulation infrared reflection absorption spectroscopy (PM-IRRAS) measurements did not yield noticeable vibrational signals, attributed to the flat-lying adsorption geometry of CDCs [90], which due to the selection rules of PM-IRRAS hindered the IR detection of CDCs. IR signatures were probed for NHCs with bulkier substituents, such as isopropyl wingtips [91, 87, 33] as they adopted an adsorption geometry perpendicular to the surface.

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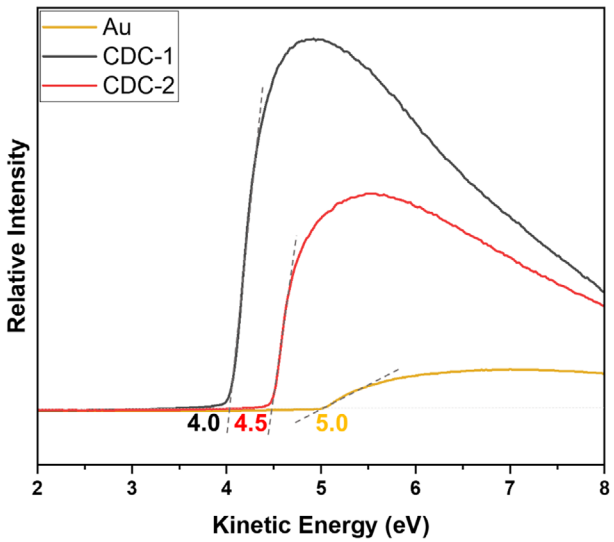


FIGURE 9 | UPS spectra of pristine Au(111) surface (yellow-colored) and following its coating with CDC-1 and CDC-2 monolayers (black and red colored spectra, respectively).

The work-function of the pristine Au(111) substrate was 4.98 eV, in agreement with previously reported values [92, 93]. The work-function values decreased to 3.99 eV and 4.45 eV following self-assembly of CDC-1 and CDC-2, respectively. This trend reflects the higher σ-donor characteristic of CDC-1 (Q_T = 0.88 and 0.70 e, for CDC-1 and CDC-2, respectively). For comparison, the work function of dimethyl-imidazole on Au(111) was measured and resulted in a value of 4.30 eV. Thus, CDC-1 lowered the work function of Au(111) by 0.99 eV, while its NHC analog lowered the work function by 0.68 eV (Figure S31).

The pronounced lowering of the work function by CDC-1 is attributed to its electron-rich nature, which enhances the electron density at the central carbene carbon (C(0)). Although the surface coverage of CDC-1 is lower than that of dimethyl-NHC due to the larger footprint of CDC, the extent of work function reduction is significantly greater, underscoring the intrinsic donor strength of CDC-1 ligand. Collectively, the experimental findings, which are supported by theoretical calculations of the charge distribution, highlight the capability of CDCs to effectively modulate the electronic structure of the gold surface, offering tunable platforms for surface functionalization.

To further investigate the chemical stability of CDC monolayers on Au film, CDC-2 coated Au films were subjected to prolonged exposure at pH 2 and pH 12 for 5 days under a nitrogen atmosphere at ambient conditions (Figure S32). The structural integrity of the monolayer was assessed using XPS measurements. Based on N1s XPS analysis, it was identified that the CDC monolayers maintained its stability under both acidic and basic conditions, showing minimal degradation. A slight shift in the N1s and C1s binding energies toward higher values was observed, which may be attributed to partial oxidation of the monolayer. These results further highlight the robustness of CDC ligands, even under exposure to harsh chemical environments. The high thermal and chemical stability of CDCs on Au highlights the crucial impact of the strong CDC-Au interactions on the surface affinity of CDC.

3 | Conclusions

This work introduces CDCs as a new class of surface ligands and establishes their ability to function as thermally robust ligands on Au(111) that significantly modify the electronic properties of the Au surface. Bench-stable hydrogen-carbonate precursors were deprotonated either in vacuo or in solution, inducing two representative surface ligands, CDC-1 and CDC-2, on Au(111). High-resolution XPS and polarized NEXAFS show that both molecules form thermally stable monolayers with a preferred flat-lying geometry upon annealing. STM imaging of CDC-2 corroborates this motif, revealing preference to adsorption on elbow sites and the transition from tilted into flat-lying geometry upon annealing to elevated temperatures [94]. The preferential adsorption of CDC-2 at Au(111) elbow sites suggests strong site-specific binding that limits the lateral diffusion required for long-range ordered self-assembly. This kinetic trapping, together with the structural flexibility of CDCs and their larger molecular footprint, in comparison to NHCs, favors a disordered rather than an ordered monolayer. DFT calculations identified the strong adsorption energies of CDCs (−3.28 eV and −2.63 eV for CDC-1 and CDC-2, respectively) and substantial charge transfer to the metal surface (0.88 e and 0.70 e for CDC-1 and CDC-2, respectively), outperforming that of NHCs. UPS measurements provided experimental validation for the dominant electronic impact of CDC adsorption on Au surface properties. Work-function reductions of 0.5–1.0 eV were measured following surface anchoring of CDCs, out-performing benchmark NHCs despite the lower surface density of CDCs. Collectively the presented data positions CDC as a strong σ-donating surface ligand, which is characterized with exceptional chemical and thermal resilience. The ability of CDC to couple large interfacial dipoles with high chemical robustness makes CDC an attractive surface ligand and open a path for leveraging the rich coordination chemistry of carbon(0) ligands in surface chemistry.

Author Contributions

Rajarshi Mondal: investigation, validation, visualization, data curation, synthesis, writing – original draft, writing – review and editing, conceptualization, methodology. **Joshua M. Sims**: investigation, validation, visualization, data curation. **Soumyajit Rajak**: investigation, validation, visualization, data curation, writing – review and editing. **Iris Berg**: writing – original draft, writing – review and editing, investigation, validation, visualization, data curation. **Shiraz Marigosian**: investigation, validation, data curation, writing – review and editing. **Luca Schio**: writing – original draft, writing – review and editing, investigation, validation, visualization, methodology, data curation. **Masoumeh Alihosseini**: investigation, validation, visualization, data curation, software. **Tzipora Ben-tzvi**: investigation, validation. **Guido Fratesi**: investigation, validation, supervision, funding acquisition, writing – original draft, writing – review and editing. **Luca Floreano**: writing – original draft, writing – review and editing, funding acquisition, investigation, validation, supervision. **Nan Jiang**: investigation, validation, supervision, writing – original draft, writing – review and editing, funding acquisition. **Carine Michel**: investigation, validation, software, supervision, writing – original draft, writing – review and editing. **Elad Gross**: writing –

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original draft, writing – review and editing, investigation, validation, conceptualization, supervision, funding acquisition.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Accession Codes

Deposition Number CCDC 2475584 contains the crystallographic data for this paper. The DFT raw data can be visualized on nomad-lab.eu, doi.org/10.17172/nomad.9ns4-y33x.

References

1. R. Tonner and G. Frenking, "C(NHC) 2: Divalent Carbon(0) Compounds with N-Heterocyclic Carbene Ligands—Theoretical Evidence for a Class of Molecules with Promising Chemical Properties," *Angewandte Chemie International Edition* 46, no. 45 (2007): 8695–8698, https://doi.org/10.1002/anie.200701632.

2. C. A. Dyker, V. Lavallo, B. Donnadieu, and G. Bertrand, "Synthesis of an Extremely Bent Acyclic Allene (A "Carbodicarbene"): A Strong Donor Ligand," *Angewandte Chemie International Edition* 47, no. 17 (2008): 3206–3209, https://doi.org/10.1002/anie.200705620.

3. A. Fürstner, M. Alcarazo, R. Goddard, and C. W. Lehmann, "Coordination Chemistry of Ene-1,1-diamines and a Prototype "Carbodicarbene"," *Angewandte Chemie International Edition* 47, no. 17 (2008): 3210–3214, https://doi.org/10.1002/anie.200705798.

4. M. A. L. Johansen and A. Ghosh, "The Curious Chemistry of Carbones," *Nature Chemistry* 15, no. 7: 1042–1042, https://doi.org/10.1038/s41557-023-01241-x.

5. S. Klein, R. Tonner, and G. Frenking, "Carbodicarbenes and Related Divalent Carbon(0) Compounds," *Chemistry – A European Journal* 16, no. 33 (2010): 10160–10170, https://doi.org/10.1002/chem.201000174.

6. M. Alcarazo, C. W. Lehmann, A. Anoop, W. Thiel, and A. Fürstner, "Coordination Chemistry at Carbon," *Nature Chemistry* 1, no. 4 (2009): 295–301, https://doi.org/10.1038/nchem.248.

7. S. Díez-González and S. P. Nolan, "Stereoelectronic Parameters Associated with N-heterocyclic carbene (NHC) Ligands: A Quest for Understanding," *Coordination Chemistry Reviews* 251, no. 5–6 (2007): 874–883, https://doi.org/10.1016/j.ccr.2006.10.004.

8. M. N. Hopkinson, C. Richter, M. Schedler, and F. Glorius, "An Overview of N-Heterocyclic Carbenes," *Nature* 510, no. 7506 (2014): 485–496, https://doi.org/10.1038/nature13384.

9. H. Jacobsen, A. Correa, A. Poater, C. Costabile, and L. Cavallo, "Understanding the M (NHC) (NHC = N-heterocyclic carbene) Bond," *Coordination Chemistry Reviews* 253, no. 5–6 (2009): 687–703, https://doi.org/10.1016/j.ccr.2008.06.006.

10. R. Tonner and G. Frenking, "Tolman's Electronic Parameters for Divalent Carbon(0) Compounds," *Organometallics* 28, no. 13 (2009): 3901–3905, https://doi.org/10.1021/om900206w.

11. W.-C. Chen, Y.-C. Hsu, C.-Y. Lee, G. P. A. Yap, and T.-G. Ong, "Synthetic Modification of Acyclic Bent Allenes (Carbodicarbenes) and Further Studies on Their Structural Implications and Reactivities," *Organometallics* 32, no. 8 (2013): 2435–2442, https://doi.org/10.1021/om400139s.

12. J. E. Walley, G. Breiner, G. Wang, and R. J. Gilliard, Jr., "s-Block Carbodicarbene Chemistry: C(sp 3)–H Activation and Cyclization Mediated by a Beryllium Center," *Chemical Communications* 55, no. 13 (2019): 1967–1970, https://doi.org/10.1039/C8CC10022E.

13. J. E. Walley, D. A. Dickie, and R. J. Gilliard, "Crystallographic Study of a Heteroleptic Chloroberyllium Borohydride Carbodicarbene Complex," *Zeitschrift für Naturforschung B* 75, no. 5 (2020): 497–501, https://doi.org/10.1515/znb-2020-0033.

14. S. Chan, P. Gupta, X. Engelmann, et al., "Observation of Carbodicarbene Ligand Redox Noninnocence in Highly Oxidized Iron Complexes," *Angewandte Chemie International Edition* 57, no. 48 (2018): 15717–15722, https://doi.org/10.1002/anie.201809158.

15. S.-C. Chan, Z. Z. Ang, P. Gupta, et al., "Carbodicarbene Ligand Redox Noninnocence in Highly Oxidized Chromium and Cobalt Complexes," *Inorganic Chemistry* 59, no. 6 (2020): 4118–4128, https://doi.org/10.1021/acs.inorgchem.0c00153.

16. R. Dolai, R. Kumar, B. J. Elvers, et al., "Carbodicarbenes and Striking Redox Transitions of Their Conjugate Acids: Influence of NHC versus CAAC as Donor Substituents," *Chemistry – A European Journal* 29, no. 2 (2023): 202202888, https://doi.org/10.1002/chem.202202888.

17. C. Yu, K. Au-Yeung, R. Liu, et al., "Diversification of the Carbodicarbene Class by Embedding an Anionic Component in Its Scaffold," *Chemistry – A European Journal* 29, no. 71 (2023): 202302886, https://doi.org/10.1002/chem.202302886.

18. N. Đorđević, R. Ganguly, M. Petković, and D. Vidović, "Bis(Carbodicarbene)Phosphenium Trication: The Case against Hypervalency," *Chemical Communications* 52, no. 63 (2016): 9789–9792, https://doi.org/10.1039/C6CC04161B.

19. K. K. Hollister, A. Molino, G. Breiner, et al., "Air-Stable Thermoluminescent Carbodicarbene-Borafluorenium Ions," *Journal of the American Chemical Society* 144, no. 1 (2022): 590–598, https://doi.org/10.1021/jacs.1c11861.

20. C.-L. Deng, A. D. Obi, B. Y. E. Tra, S. K. Sarkar, D. A. Dickie, and R. J. Gilliard, "Air- and Photo-Stable Luminescent Carbodicarbene-Azaboraacenium Ions," *Nature Chemistry* 16, no. 3 (2024): 437–445, https://doi.org/10.1038/s41557-023-01381-0.

21. T. Shigeta, S. Takano, and T. Tsukuda, "Carbodiphosphorane as an Electron-Donating Ligand for Gold Nanoclusters," *Langmuir* 41, no. 25 (2025): 16180–16186, https://doi.org/10.1021/acs.langmuir.5c01355.

22. Y. Chan, Y. Bai, W. Chen, et al., "Synergistic Catalysis by Brønsted Acid/Carbodicarbene Mimicking Frustrated Lewis Pair-Like Reactivity," *Angewandte Chemie International Edition* 60, no. 36 (2021): 19949–19956, https://doi.org/10.1002/anie.202107127.

23. B. Huang, G. S. Sontakke, Y. Cheng, et al., "Frustrated Lewis Pair (FLP) Reactivity from Carbone–BPh 3 Lewis Adduct," *Chemistry – A European Journal* 31 (2025): 02344, https://doi.org/10.1002/chem.202502344.

24. W.-C. Chen, W.-C. Shih, T. Jurca, et al., "Carbodicarbenes: Unexpected π-Accepting Ability during Reactivity with Small Molecules," *Journal of the American Chemical Society* 139, no. 36 (2017): 12830–12836, https://doi.org/10.1021/jacs.7b08031.

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25. Y. He, Y. Lyu, D. Tymann, P. W. Antoni, and M. M. Hansmann, "Cleavage of Carbodicarbenes with N_2O for Accessing Stable Diazoalkenes: Two-Fold Ligand Exchange at a C(0)-Atom," *Angewandte Chemie International Edition* 64, no. 3 (2025): 202415228, https://doi.org/10.1002/anie.202415228.

26. F. Krischer and V. H. L. Gessner, "Ligand Exchange at Carbon: Synthetic Entry to Elusive Species and Versatile Reagents," *JACS Au* 4, no. 5 (2024): 1709–1722, https://doi.org/10.1021/jacsau.4c00112.

27. A. Das, Q. Le Dé, and V. H. Gessner, "Rethinking Carbanion Chemistry from Donor Substituents to Weakly Coordinating Carbanions," *Nature Reviews Chemistry* 9 (2025): 523–536, https://doi.org/10.1038/s41570-025-00725-3.

28. S. Liu, W. Shih, W. Chen, and T. Ong, "Carbodicarbenes and Their Captodative Behavior in Catalysis," *Chemcatchem* 10, no. 7 (2018): 1483–1498, https://doi.org/10.1002/cctc.201701577.

29. W.-C. Chen, C.-Y. Lee, B.-C. Lin, et al., "The Elusive Three-Coordinate Dicationic Hydrido Boron Complex," *Journal of the American Chemical Society* 136, no. 3 (2014): 914–917, https://doi.org/10.1021/ja4120852.

30. J. E. Walley, L. S. Warring, G. Wang, et al., "Carbodicarbene Bismaalkene Cations: Unravelling the Complexities of Carbene versus Carbone in Heavy Pnictogen Chemistry," *Angewandte Chemie International Edition* 60, no. 12 (2021): 6682–6690, https://doi.org/10.1002/anie.202014398.

31. A. V. Zhukhovitskiy, M. G. Mavros, T. Van Voorhis, and J. A. Johnson, "Addressable Carbene Anchors for Gold Surfaces," *Journal of the American Chemical Society* 135, no. 20 (2013): 7418–7421, https://doi.org/10.1021/ja401965d.

32. C. M. Crudden, J. H. Horton, I. I. Ebralidze, et al., "Ultra Stable Self-Assembled Monolayers of N-Heterocyclic Carbenes on Gold," *Nature Chemistry* 6, no. 5 (2014): 409–414, https://doi.org/10.1038/nchem.1891.

33. G. Wang, A. Rühling, S. Amirjalayer, et al., "Ballbot-Type Motion of N-Heterocyclic Carbenes on Gold Surfaces," *Nature Chemistry* 9, no. 2 (2017): 152–156, https://doi.org/10.1038/nchem.2622.

34. E. Amit, L. Dery, S. Dery, et al., "Electrochemical Deposition of N-Heterocyclic Carbene Monolayers on Metal Surfaces," *Nature Communications* 11, no. 1 (2020): 5714, https://doi.org/10.1038/s41467-020-19500-7.

35. M. Koy, P. Bellotti, M. Das, and F. Glorius, "N-Heterocyclic Carbenes as Tunable Ligands for Catalytic Metal Surfaces," *Nature Catalysis* 4, no. 5 (2021): 352–363, https://doi.org/10.1038/s41929-021-00607-z.

36. I. Berg, R. Mondal, J. M. Sims, et al., "Strong Substrate–Adsorbate Interactions Direct the Impact of Fluorinated N-Heterocyclic Carbene Monolayers on Au Surface Properties," *ACS Applied Materials & Interfaces* 16, no. 47 (2024): 65469–65479, https://doi.org/10.1021/acsami.4c12514.

37. A. Das, T. Hung, A. Rank, et al., "A Free N-Heterocyclic Carbene and Its Metal Complex," *Angewandte Chemie International Edition* 65, no. 19 (2026): 24266, https://doi.org/10.1002/anie.202524266.

38. I. Berg, L. Schio, J. Reitz, et al., "Self-Assembled Monolayers of N-Heterocyclic Olefins on Au(111)," *Angewandte Chemie International Edition* 62, no. 46 (2023): 202311832, https://doi.org/10.1002/anie.202311832.

39. M. Das, C. Hogan, R. Zielinski, et al., "N-Heterocyclic Olefins on a Silicon Surface," *Angewandte Chemie International Edition* 62, no. 50 (2023): 202314663, https://doi.org/10.1002/anie.202314663.

40. F. Landwehr, M. Das, S. Tosoni, et al., "N-Heterocyclic Olefins on a Metallic Surface – Adsorption, Orientation, and Electronic Influence," *Advanced Materials Interfaces* 11, no. 31 (2024): 2400378, https://doi.org/10.1002/admi.202400378.

41. F. Landwehr, A. Das, S. Tosoni, et al., "Writing with Molecules: Tip-Induced Local Chemisorption of N-Heterocyclic Olefins on Cu(111)," *Journal of the American Chemical Society* 147, no. 31 (2025): 27676–27684, https://doi.org/10.1021/jacs.5c06188.

42. A. Bakker, A. Timmer, E. Kolodzeiski, et al., "Elucidating the Binding Modes of N-Heterocyclic Carbenes on a Gold Surface," *Journal of the American Chemical Society* 140, no. 38 (2018): 11889–11892, https://doi.org/10.1021/jacs.8b06180.

43. C. R. Larrea, C. J. Baddeley, M. R. Narouz, N. J. Mosey, J. H. Horton, and C. M. Crudden, "N -Heterocyclic Carbene Self-assembled Monolayers on Copper and Gold: Dramatic Effect of Wingtip Groups on Binding, Orientation and Assembly," *Chemphyschem* 18, no. 24 (2017): 3536–3539, https://doi.org/10.1002/cphc.201701045.

44. E. Amit, R. Mondal, I. Berg, Z. Nairoukh, and E. Gross, "N-Heterocyclic Carbene Monolayers on Metal-Oxide Films: Correlations between Adsorption Mode and Surface Functionality," *Langmuir* 40, no. 19 (2024): 10374–10383, https://doi.org/10.1021/acs.langmuir.4c01109.

45. G. Kaur, R. L. Thimes, J. P. Camden, and D. Jenkins, "Fundamentals and Applications of N-Heterocyclic Carbene Functionalized Gold Surfaces and Nanoparticles," *Chemical Communications* 58, no. 95 (2022): 13188–13197, https://doi.org/10.1039/D2CC05183D.

46. Y. Pan, A. Das, F. Glorius, and J. Ren, "Insights into the Surface Chemistry of N-Heterocyclic Carbenes," *Chemical Society Reviews* 54 (2025): 4626–4650, https://doi.org/10.1039/D4CS01299B.

47. S. Kang, S. E. Byeon, and H. J. Yoon, "N -Heterocyclic Carbene Anchors in Electronics Applications," *Bulletin of the Korean Chemical Society* 42, no. 5: 712–723, https://doi.org/10.1002/bkcs.12261.

48. C. M. Crudden, J. H. Horton, M. R. Narouz, et al., "Simple Direct Formation of Self-Assembled N-Heterocyclic Carbene Monolayers on Gold and Their Application in Biosensing," *Nature Communications* 7, no. 1 (2016): 12654, https://doi.org/10.1038/ncomms12654.

49. J. Fan, S. Kang, and M. Gupta, "Long-Term Stability of N-Heterocyclic Carbene (NHC) Functionalized Organic Electrochemical Transistor (OECT) for Biosensing Applications," *Frontiers in Sensors* 5 (2024): 1430958, https://doi.org/10.3389/fsens.2024.1430958.

50. Z. Li, K. Munro, I. I. Ebralize, et al., "N-Heterocyclic Carbene Self-Assembled Monolayers on Gold as Surface Plasmon Resonance Biosensors," *Langmuir* 33 (2017): 13936–13944, https://doi.org/10.1021/acs.langmuir.7b03280.

51. S. Roland, X. Ling, and M.-P. N.-H. C. Pileni, "N -Heterocyclic Carbene Ligands for Au Nanocrystal Stabilization and Three-Dimensional Self-Assembly," *Langmuir* 32, no. 31 (2016): 7683–7696, https://doi.org/10.1021/acs.langmuir.6b01458.

52. M. Ghosh and S. Khan, "N-Heterocyclic Carbenes Capped Metal Nanoparticles: An Overview of Their Catalytic Scope," *ACS Catalysis* 13, no. 14 (2023): 9313–9325, https://doi.org/10.1021/acscatal.3c01824.

53. V. K. Kulkarni, E. L. Albright, E. Zeinizade, et al., "Impact of Ligand Structure on Biological Activity and Photophysical Properties of NHC-Protected Au13 Nanoclusters," *Journal of the American Chemical Society* 147, no. 5 (2025): 4017–4025, https://doi.org/10.1021/jacs.4c12072.

54. J. F. DeJesus, L. M. Sherman, D. J. Yohannan, et al., "A Benchtop Method for Appending Protic Functional Groups to N-Heterocyclic Carbene Protected Gold Nanoparticles," *Angewandte Chemie International Edition* 59, no. 19 (2020): 7585–7590, https://doi.org/10.1002/anie.202001440.

55. I. M. Jensen, S. Chowdhury, G. Hu, L. Jensen, J. P. Camden, and D. M. Jenkins, "Seeking a Au–C stretch on gold nanoparticles with 13 C-labeled N-heterocyclic carbenes," *Chemical Communications* 59, no. 98 (2023): 14524–14527, https://doi.org/10.1039/D3CC04973F.

56. Q.-A. Huang, M. Takaki, H. Murayama, et al., "Supported Gold Nanoparticles Prepared from NHC-Au Complex Precursors as Reusable Heterogeneous Catalysts," *Molecular Catalysis* 549 (2023): 113460, https://doi.org/10.1016/j.mcat.2023.113460.

57. R. W. Y. Man, H. Yi, S. Malola, et al., "Synthesis and Characterization of Enantiopure Chiral Bis NHC-Stabilized Edge-Shared Au10 Nanocluster with Unique Prolate Shape," *Journal of the American Chemical Society* 144, no. 5 (2022): 2056–2061, https://doi.org/10.1021/jacs.1c11857.

58. R. W. Y. Man, C.-H. Li, M. W. A. MacLean, et al., "Ultrastable Gold Nanoparticles Modified by Bidentate N -Heterocyclic Carbene Ligands,"

16136829, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/smll.76364 by Joshua M. Sims - Ku Leuven , Wiley Online Library on [24/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Journal of the American Chemical Society 140, no. 5 (2018): 1576–1579, https://doi.org/10.1021/jacs.7b08516.

59. K. Salorinne, R. W. Y. Man, C. Li, M. Taki, M. Nambo, and C. M. Crudden, "Water-Soluble N-Heterocyclic Carbene-Protected Gold Nanoparticles: Size-Controlled Synthesis, Stability, and Optical Properties," *Angewandte Chemie International Edition* 56, no. 22 (2017): 6198–6202, https://doi.org/10.1002/anie.201701605.

60. J. Ren, M. Koy, H. Osthues, et al., "On-Surface Synthesis of Ballbot-Type N-Heterocyclic Carbene Polymers," *Nature Chemistry* 15, no. 12 (2023): 1737–1744, https://doi.org/10.1038/s41557-023-01310-1.

61. S. Dery, H. Mehlman, L. Hale, et al., "Site-Independent Hydrogenation Reactions on Oxide-Supported Au Nanoparticles Facilitated by Intraparticle Hydrogen Atom Diffusion," *ACS Catalysis* 11, no. 15 (2021): 9875–9884, https://doi.org/10.1021/acscatal.1c01987.

62. I. Berg, L. Hale, M. Carmiel-Kostan, F. D. Toste, and E. Gross, "Using Silyl Protecting Group to Enable Post-Deposition C–C Coupling Reactions of Alkyne-Functionalized N-Heterocyclic Carbene Monolayers on Au Surfaces," *Chemical Communications* 57, no. 43 (2021): 5342–5345, https://doi.org/10.1039/D1CC01271A.

63. E. A. Doud, M. S. Inkpen, G. Lovat, et al., "In Situ Formation of N-Heterocyclic Carbene-Bound Single-Molecule Junctions," *Journal of the American Chemical Society* 140, no. 28 (2018): 8944–8949, https://doi.org/10.1021/jacs.8b05184.

64. M. Das, M. Kohlstädt, M. Enders, et al., "Surface Modification of ITO with N-Heterocyclic Carbene Precursors Results in Electron Selective Contacts in Organic Photovoltaic Devices," *Chemistry – A European Journal* 29, no. 60 (2023): 202301482, https://doi.org/10.1002/chem.202301482.

65. A. Lv, M. Freitag, K. M. Chepiga, A. H. Schäfer, F. Glorius, and L. Chi, "N-Heterocyclic-Carbene-Treated Gold Surfaces in Pentacene Organic Field-Effect Transistors: Improved Stability and Contact at the Interface," *Angewandte Chemie International Edition* 57, no. 17 (2018): 4792–4796, https://doi.org/10.1002/anie.201713415.

66. Z. Wang, M. Das, C. Gutheil, et al., "Surface Modification with a Fluorinated N-Heterocyclic Carbene on Au: Effect on Contact Resistance in Organic Field-Effect Transistors," *Journal of Materials Chemistry C* 10, no. 22 (2022): 8589–8595, https://doi.org/10.1039/D2TC01311H.

67. S. Park, S. Kang, and H. J. Yoon, "Thermopower of Molecular Junction in Harsh Thermal Environments," *Nano Letters* 22, no. 10 (2022): 3953–3960, https://doi.org/10.1021/acs.nanolett.2c00422.

68. A. Das, A. Borrini, C. Gutheil, et al., "Stable Synapse-Like Memory Switching in N-Heterocyclic Carbene Monolayers," *Angewandte Chemie International Edition* 65, no. 25 (2026): 1823213, https://doi.org/10.1002/anie.1823213.

69. S. Ikemoto, S. Muratsugu, T. Koitaya, et al., "Coordination-Induced Trigger for Activity: N-Heterocyclic Carbene-Decorated Ceria Catalysts Incorporating Cr and Rh with Activity Induction by Surface Adsorption Site Control," *Journal of the American Chemical Society* 145, no. 3 (2023): 1497–1504, https://doi.org/10.1021/jacs.2c07290.

70. R. J. Lewis, M. Koy, M. Macino, et al., "N-Heterocyclic Carbene Modified Palladium Catalysts for the Direct Synthesis of Hydrogen Peroxide," *Journal of the American Chemical Society* 144, no. 34 (2022): 15431–15436, https://doi.org/10.1021/jacs.2c04828.

71. M. G. Avello, S. Golling, L. Truong-Phuoc, et al., "(NHC-olefin)-nickel(0) Nanoparticles as Catalysts for the (Z)-Selective Semi-Hydrogenation of Alkynes and Ynamides," *Chemical Communications* 59, no. 11 (2023): 1537–1540, https://doi.org/10.1039/D2CC05302K.

72. J. B. Ernst, S. Muratsugu, F. Wang, M. Tada, and F. Glorius, "Tunable Heterogeneous Catalysis: N-Heterocyclic Carbenes as Ligands for Supported Heterogeneous Ru/K-Al2 O3 Catalysts To Tune Reactivity and Selectivity," *Journal of the American Chemical Society* 138, no. 34 (2016): 10718–10721, https://doi.org/10.1021/jacs.6b03821.

73. L. Amar, R. Mondal, O. Blumen, et al., "Self-Assembled Monolayer of N-Heterocyclic Carbene as a Primer in a Dual-Layer Coating for Corrosion Protection on Iron," *Angewandte Chemie* 137, no. 19 (2025): 202422879, https://doi.org/10.1002/ange.202422879.

74. L. Laundry-Mottiar, T. M. Suduwella, W. G. K. Senanayake, et al., "N-Heterocyclic Carbene Overlayers on Mild Steel," *Chemistry of Materials* 37, no. 1 (2025): 76–86, https://doi.org/10.1021/acs.chemmater.4c01804.

75. I. Berg, E. Amit, L. Hale, F. D. Toste, and E. Gross, "N -Heterocyclic Carbene Based Nanolayer for Copper Film Oxidation Mitigation," *Angewandte Chemie International Edition* 61, no. 25 (2022): 202201093, https://doi.org/10.1002/anie.202201093.

76. L. Zhang, Z. Liu, Y. Zhang, X. Xu, and Y. Tang, "Facile Formation of N-Heterocyclic Carbene Self-Assembled Monolayers and Protective Performance for Silver Substrates," *Materials Chemistry and Physics* 346 (2025): 131322, https://doi.org/10.1016/j.matchemphys.2025.131322.

77. D. T. Nguyen, M. Freitag, M. Körsgen, et al., "Versatile Micropatterns of N-Heterocyclic Carbenes on Gold Surfaces: Increased Thermal and Pattern Stability with Enhanced Conductivity," *Angewandte Chemie International Edition* 57, no. 35 (2018): 11465–11469, https://doi.org/10.1002/anie.201807197.

78. G. Lovat, E. A. Doud, D. Lu, et al., "Determination of the Structure and Geometry of N-Heterocyclic Carbenes on Au(111) Using High-Resolution Spectroscopy," *Chemical Science* 10, no. 3 (2019): 930–935, https://doi.org/10.1039/C8SC03502D.

79. M. J. Trujillo, S. L. Strausser, J. C. Becca, et al., "Using SERS To Understand the Binding of N-Heterocyclic Carbenes to Gold Surfaces," *The Journal of Physical Chemistry Letters* 9, no. 23 (2018): 6779–6785, https://doi.org/10.1021/acs.jpclett.8b02764.

80. S. Dery, S. Kim, G. Tomaschun, et al., "Elucidating the Influence of Anchoring Geometry on the Reactivity of NO2 -Functionalized N-Heterocyclic Carbene Monolayers," *The Journal of Physical Chemistry Letters* 10, no. 17 (2019): 5099–5104, https://doi.org/10.1021/acs.jpclett.9b01808.

81. A. Das, P. Chahar, and F. Glorius, "How to Get Them on: N-Heterocyclic Carbenes (NHCs) on Surfaces," *Helvetica Chimica Acta* 109, no. 3 (2026): 00201, https://doi.org/10.1002/hlca.202500201.

82. A. Calabrese, L. Floreano, A. Verdini, C. Mariani, and M. G. Betti, "Filling Empty States in a CuPc Single Layer on the Au(110) Surface via Electron Injection," *Physical Review B* 79, no. 11 (2009): 115446, https://doi.org/10.1103/PhysRevB.79.115446.

83. L. Floreano, A. Cossaro, R. Gotter, et al., "Periodic Arrays of Cu-Phthalocyanine Chains on Au(110)," *The Journal of Physical Chemistry C* 112, no. 29 (2008): 10794–10802, https://doi.org/10.1021/jp711140e.

84. R. J. J. Jansen and H. Van Bekkum, "XPS of Nitrogen-Containing Functional Groups on Activated Carbon," *Carbon* 33, no. 8 (1995): 1021–1027, https://doi.org/10.1016/0008-6223(95)00030-H.

85. F.-M. Pan, P. C. Stair, and T. H. Fleisch, "Chemisorption of Pyridine and Pyrrole on Iron Oxide Surfaces Studied by XPS," *Surface Science* 177, no. 1 (1986): 1–13, https://doi.org/10.1016/0039-6028(86)90253-0.

86. M. Wróbel, D. M. Cegiełka, A. Asyuda, K. Kozieł, M. Zharnikov, and P. Cyganik, "N-Heterocyclic Carbenes – The Design Concept for Densely Packed and Thermally Ultra-Stable Aromatic Self-Assembled Monolayers," *Nano Today* 53 (2023): 102024, https://doi.org/10.1016/j.nantod.2023.102024.

87. M. Furlan, P. Melix, M. D. Aloisio, et al., "High-Density Self-Assembled Monolayers of N-Heterocyclic Carbenes on Au(111)—Observation of Normal and Mesoionic Bonding Modes," *Chemistry of Materials* 37, no. 15 (2025): 5753–5763, https://doi.org/10.1021/acs.chemmater.5c00886.

88. L. Jiang, B. Zhang, G. Médard, et al., "N -Heterocyclic Carbenes on Close-Packed Coinage Metal Surfaces: Bis-Carbene Metal Adatom Bonding Scheme of Monolayer Films on Au, Ag and Cu," *Chemical Science* 8, no. 12 (2017): 8301–8308, https://doi.org/10.1039/C7SC03777E.

16136829, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/smll.76364 by Joshua M. Sims - Ku Leuven , Wiley Online Library on [24/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

89. J. N. S. Stöhr, G. Ertl, R. Gomer, D. L. Mills, and H. K. V. Lotsch, *NEXAFS Spectroscopy* (Springer Berlin, Heidelberg, 1992), https://doi.org/10.1007/978-3-662-02853-7.

90. E. DesRoche, F. Crasto De Lima, R. R. K. Groome, et al., "Mechanisms of N-Heterocyclic Carbene Complex Lattice Formation," *ACS Nano* 19, no. 35 (2025): 31417–31427, https://doi.org/10.1021/acsnano.5c06139.

91. Y. Zeng, T. Zhang, M. R. Narouz, C. M. Crudden, and P. H. McBreen, "Generation and Conversion of an N-Heterocyclic Carbene on Pt(111)," *Chemical Communications* 54, no. 88 (2018): 12527–12530, https://doi.org/10.1039/C8CC06894A.

92. H. K. Kim, A. S. Hyla, P. Winget, et al., "Reduction of the Work Function of Gold by N-Heterocyclic Carbenes," *Chemistry of Materials* 29, no. 8: 3403–3411, https://doi.org/10.1021/acs.chemmater.6b04213.

93. D. A. R. Nanan, J. T. Lomax, J. Bentley, et al., "Self-Assembled Monolayers of Triazolylidenes on Gold and Mixed Gold/Dielectric Substrates," *Journal of the American Chemical Society* 147, no. 7 (2025): 5624–5631, https://doi.org/10.1021/jacs.4c11125.

94. P. Wiesener, A. Das, E. Kolodzeiski, et al., "Step-Edge Functionalization by N-Heterocyclic Carbenes Enhances Catalytic Activity in Electrochemical CO 2 Reduction," *Advanced Materials* 38, no. 39 (2026): 73621, https://doi.org/10.1002/adma.73621.

95. E. Quemener and M. Corvellec, "SIDUS—the Solution for Extreme Deduplication of an Operating System," *Linux Journal* 2013, no. 235 (2013): 3.

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